



1. CRYPTOGRAPHIC PRIMITIVES & ENCRYPTION MODEL

AEGIS cryptographic architecture relies on modern primitives with high hybrid post-quantum resistance and key sovereignty:

FUNCTION	PRIMITIVE / ALGORITHM	SPECIFICATION & KEY SIZE
Key Exchange (KEX)	X25519 (Elliptic Curve)	256-bit ECDH ephemeral keys generated in RAM
Signature & Identity	Ed25519	Non-persistent authentication key pairs
Symmetric Encryption	ChaCha20-Poly1305 / AES-256-GCM	AEAD with unique 96-bit Nonce per packet
Key Derivation (KDF)	HKDF-SHA256 / Argon2id	Dynamic entropy salts (Memory-hard)

Double-Ratchet Protocol (Forward & Break-in Secrecy)

Key chain refreshed for every sent/received message. Session key compromise has zero impact on past or future communications.

2. MEMORY HARDENING (RAM ISOLATION & ANTI-FORENSICS)

- Strict Zero-Persistence:** Host OS strictly prohibits calls to `SharedPreferences`, `SQLite`, or writing disk cache files.
- Swapping Protection:** Allocated RAM locked via `mlock()` preventing kernel from paging memory to SWAP or hibernation files.
- Memory Wipe (Zeroize):** Native Rust / C++ `Zeroize` trait implementation. Key buffers, decrypted messages, and network structures are overwritten with binary zeros (`0x00`) upon release.
- Cold Boot Defense:** Hardware timer forced purge (3 min inactivity) triggering key pointer destruction and immediate `exit(0)` invocation.

3. NATIVE ANDROID HARDENING (OS-LEVEL MITIGATIONS)

UI Attack Surface (FLAG_SECURE)

Enforced `WindowManager.LayoutParams.FLAG_SECURE` native flag. Hardware-blocks screen captures/recordings and clears display buffer in OS task switcher.

Dynamic PIN Mechanism & Anti-Coercion (DURESS & KILL)

Session password is dynamic. Entering **ANY valid PIN** other than the primary key or system codes displays a decoy interface ("Notes"). Entering panic code (e.g., `0000`) or repeated failed attempts instantly wipes RAM and terminates process.

4. NETWORK TRANSPORT ARCHITECTURE & FREQUENCY HOPPING

AEGIS features a dynamic Transport Switch with anti-censorship frequency hopping strategy:

TRANSPORT MODE	PROTOCOL STACK	IMPLEMENTATION & RESILIENCE
Embedded Tor v3	v3 Hidden Services (.onion) + Obfs4	Maximum IP-layer anonymity, circumvents national DPI firewalls.
Direct P2P WAN	Kademlia DHT + STUN / WebRTC	Distributed direct P2P routing, minimal latency without central servers.
Local Mesh (LAN/BT)	Wi-Fi Direct / Multicast UDP / Bluetooth Mesh	Completely air-gapped from Internet for physical meetings.
Frequency Hopping	Dynamic Failover + Snowflake / Pluggable Transports	Real-time dynamic channel/protocol switching upon channel degradation, DPI blocking, or network jamming.

Network Frequency Hopping Regulation

In case of outage or active state DPI inspection, AEGIS seamlessly hops from Tor v3 to WebRTC/Snowflake and Bluetooth/Wi-Fi mesh without breaking E2EE session.

5. STEGANOGRAPHY MODULE (TEXT-BASED STEGANOGRAPHY)

- Covert Encoding:** Conceals public keys and encrypted payloads inside natural language text structures (poems, prose).
- Utf-8 Zero-Width Mapping:** Converts encrypted binary bytes into invisible Unicode characters (Zero-Width Space `U+200B`, Zero-Width Non-Joiner `U+200C`) interleaved in carrier text.
- Entropy-Resistant Analysis:** Bypasses content inspection scanners (DPI) and word-frequency statistical steganalysis.

6. SECURITY AUDIT & COMPLIANCE MATRIX

THREAT VECTOR	APPLIED TECHNICAL MITIGATION	COMPLIANCE STATUS
Physical Seizure of Unlocked Phone	Dynamic DURESS (Decoy), Kill Switch (`0000`) & 3-min Timer	COMPLIANT
Post-Mortem Disk Forensic Analysis (NAND)	Zero-Persistence Architecture (0 bytes written)	COMPLIANT
Global Network Interception / Eavesdropping	Network Frequency Hopping, Tor v3 Routing + Double-Ratchet E2EE	COMPLIANT
Cold Boot RAM Memory Grab Attack	`mlock` invocations & Automatic Rust Zeroize	COMPLIANT

Audit Resources & Ares Maxima Official Documentation:

<https://en.aresmaxima.com/aegis-p2p/>